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JANUARY 2026

HTTP Log Analysis

Splunk Basics – HTTP Log Analysis Lab Report

STRUCTURED LAB REPORT

OBJECTIVE

The objective of this lab is to learn how **HTTP web traffic logs can be analyzed using Splunk** in order to detect web-based attacks and suspicious activity.

During this lab we will:

- Upload **Zeek-style HTTP logs** into Splunk.
- Analyze **web traffic generated by clients**.
- Detect **server errors and suspicious user agents**.
- Identify **large file downloads or uploads**.

HTTP log analysis helps security teams detect:

- Web scanning tools
- Automated attack scripts
- Data exfiltration through large transfers
- Suspicious web access patterns

TOOLS AND ENVIRONMENT

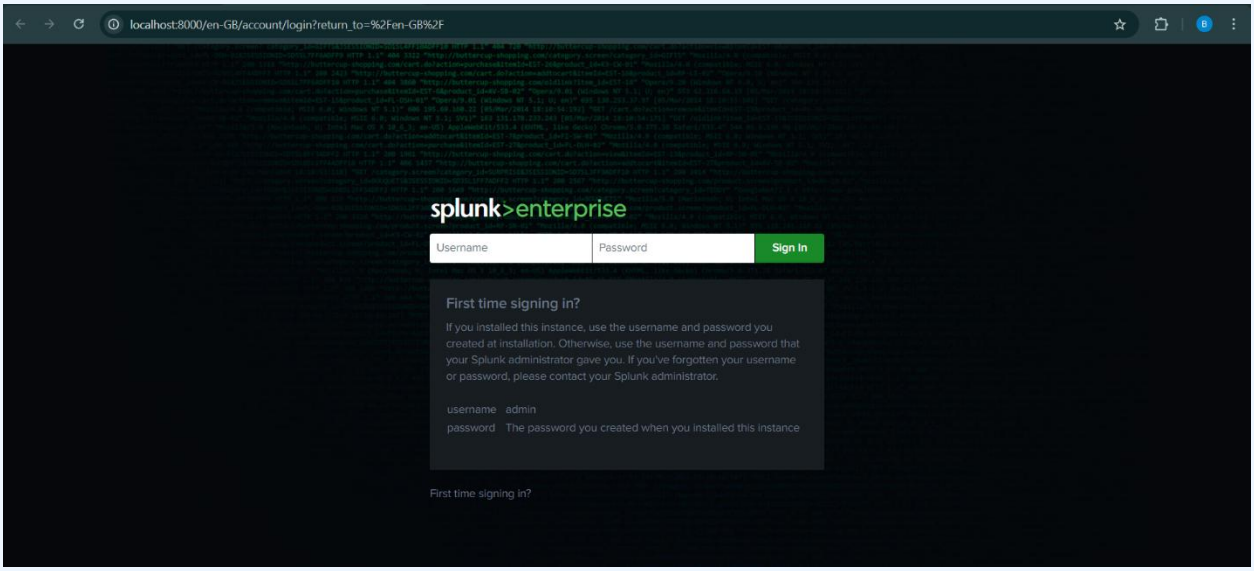
Component	Description
Splunk	Log monitoring and data analysis platform
Data Source	Zeek HTTP logs
Log Format	JSON
Index Used	http_lab
Sourcetype	json

LAB SETUP

Before starting the analysis, the HTTP log file must be uploaded into Splunk.

Step 1 – Open Splunk Web

1. Open **Splunk Web Interface** in your browser.
2. Log in using your credentials.

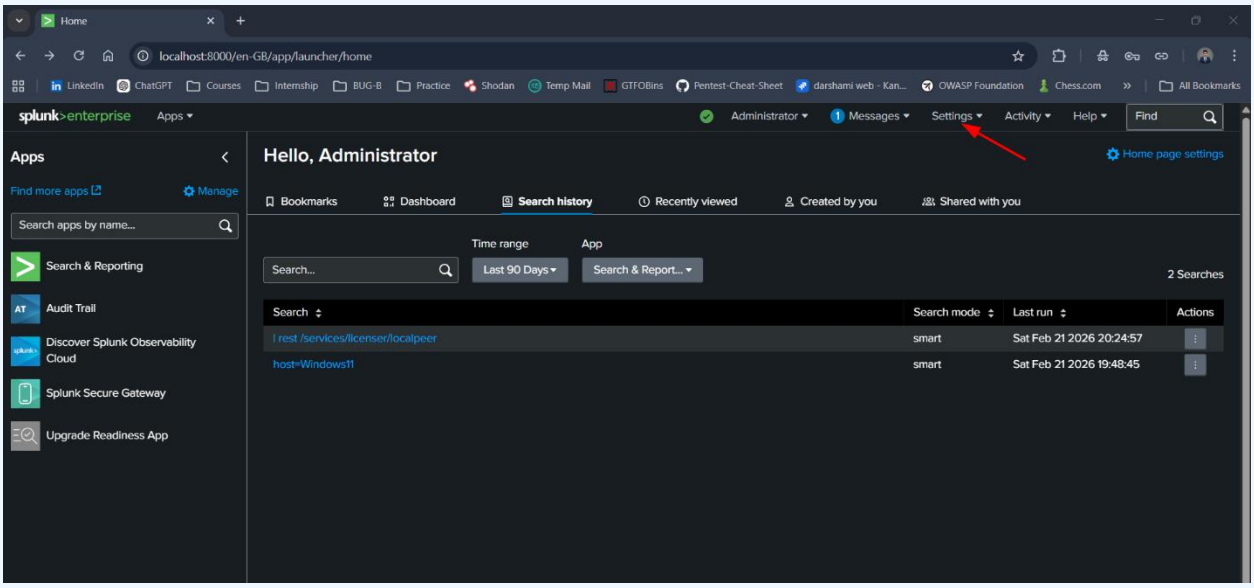


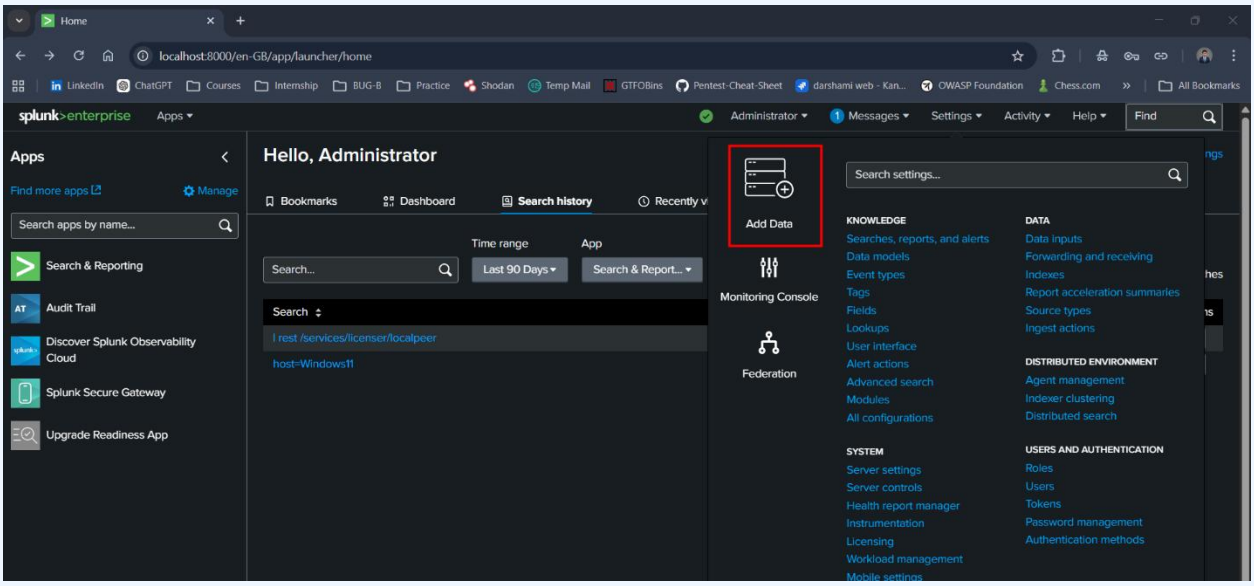
Step 2 – Navigate to Add Data

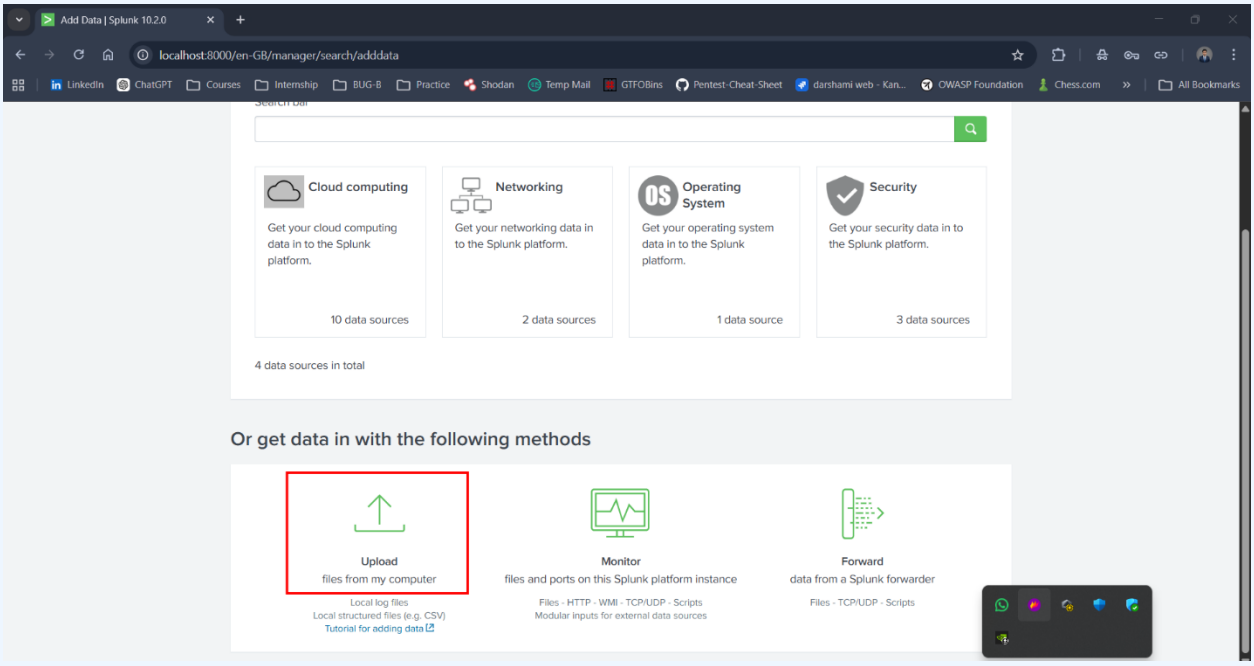
1. Go to:

Settings → Add Data

2. Select **Upload**.





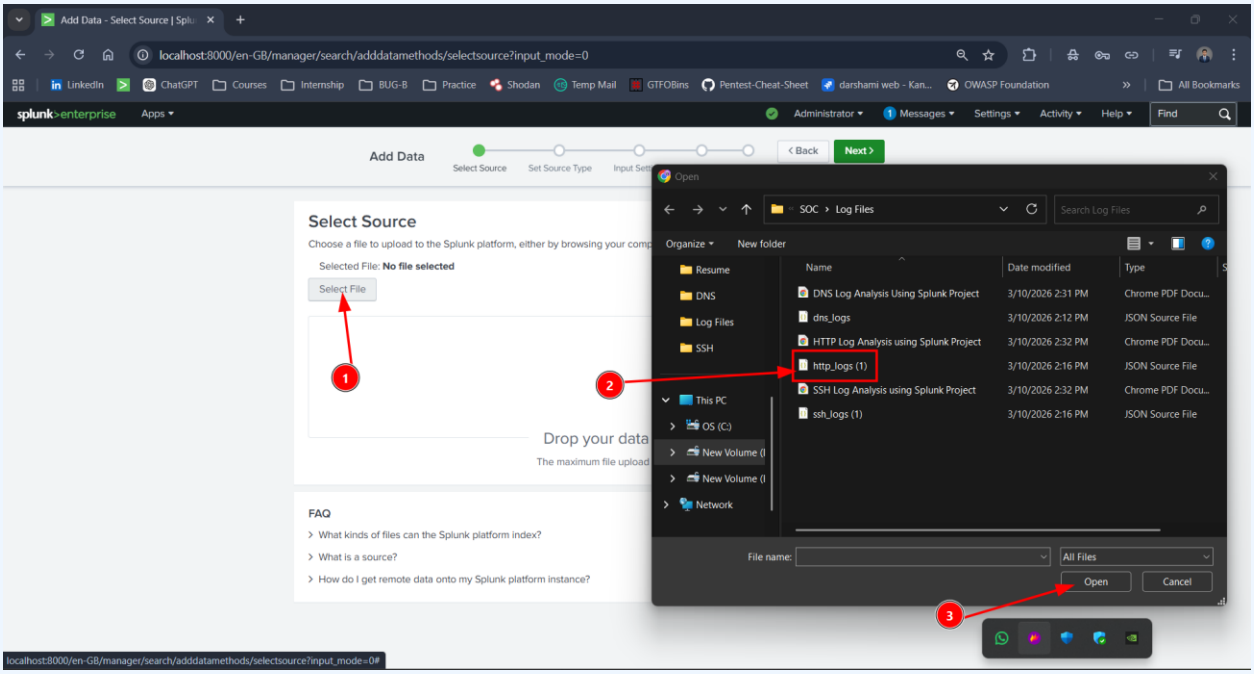


Step 3 – Upload the HTTP Log File

1. Click **Select File**
2. Upload the file:

```
http_logs.json
```

This dataset contains **HTTP requests and responses recorded by Zeek network monitoring tool**.



Step 4 – Configure Source Type

Set the source type to:

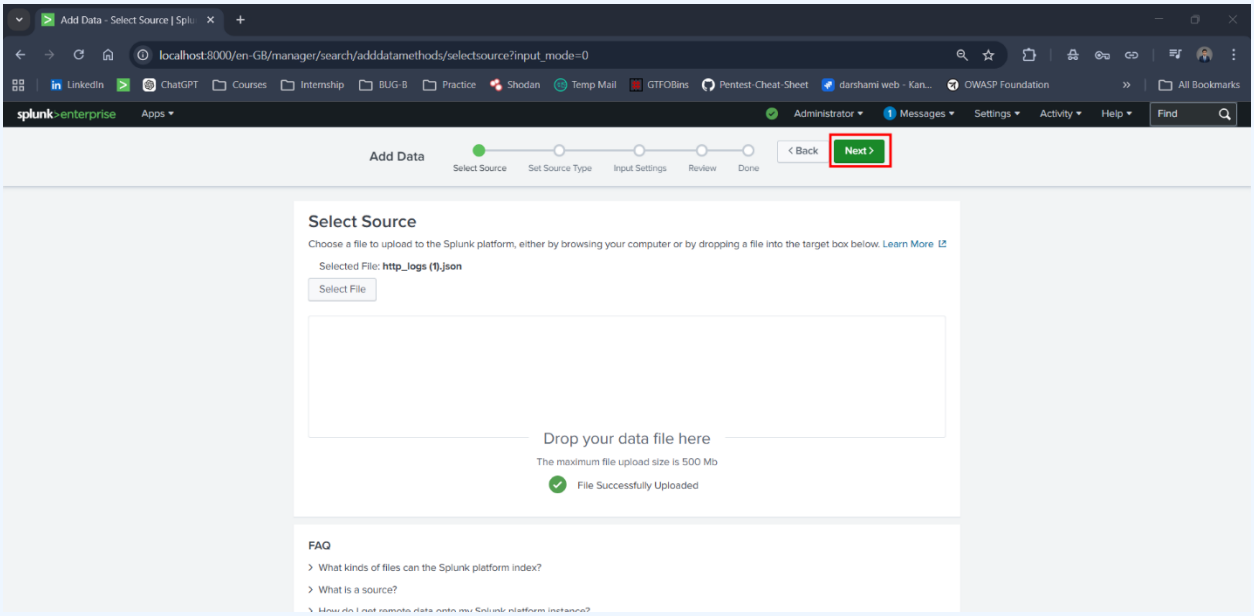
```
_json
```

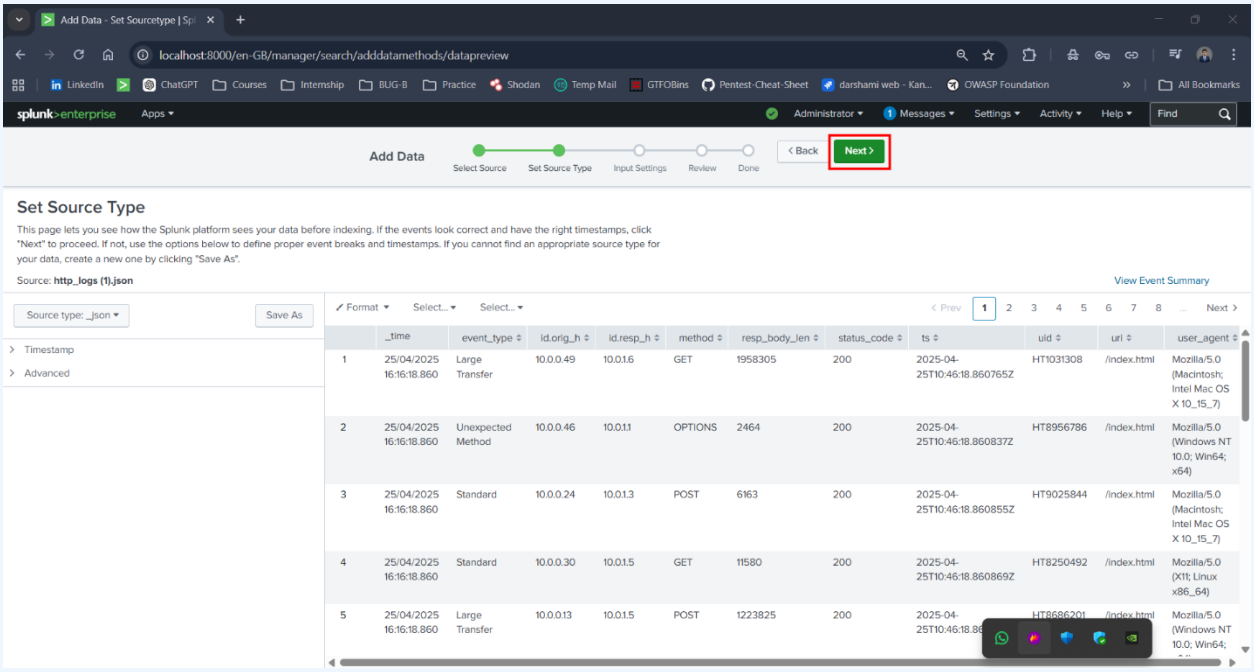
or create a custom source type:

```
http_logs
```

Reason:

- The logs are structured in **JSON format**
- Splunk automatically extracts important fields like:
  - id.orig_h
  - id.resp_h
  - uri
  - user_agent
  - status_code
  - resp_body_len





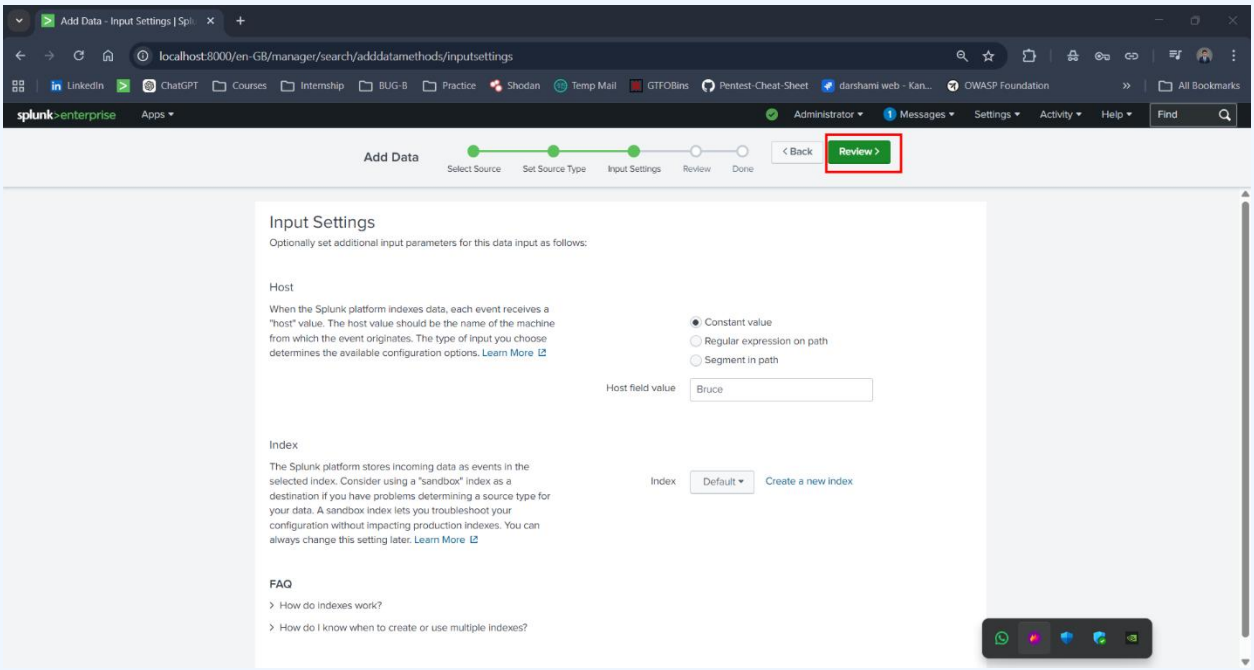
Step 5 – Configure Index

Select or create the index:

```
http_lab
```

Reason:

- Keeps HTTP analysis data separate from other logs.
- Makes searching faster and easier.



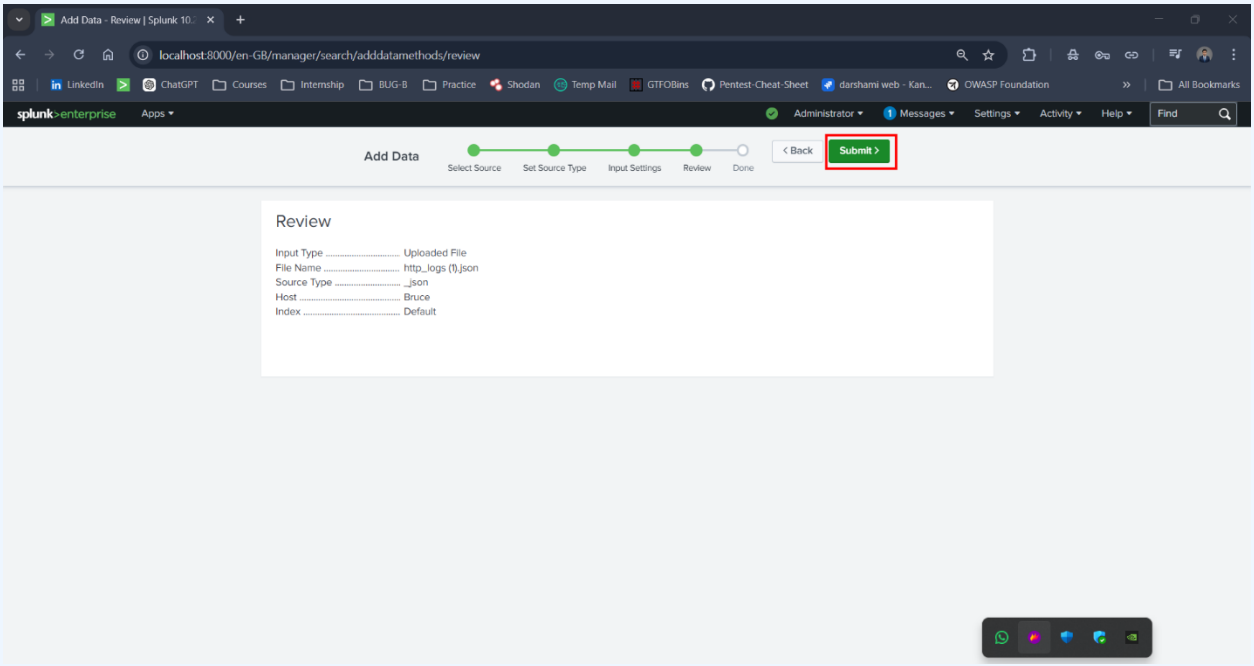
Step 6 – Confirm Upload

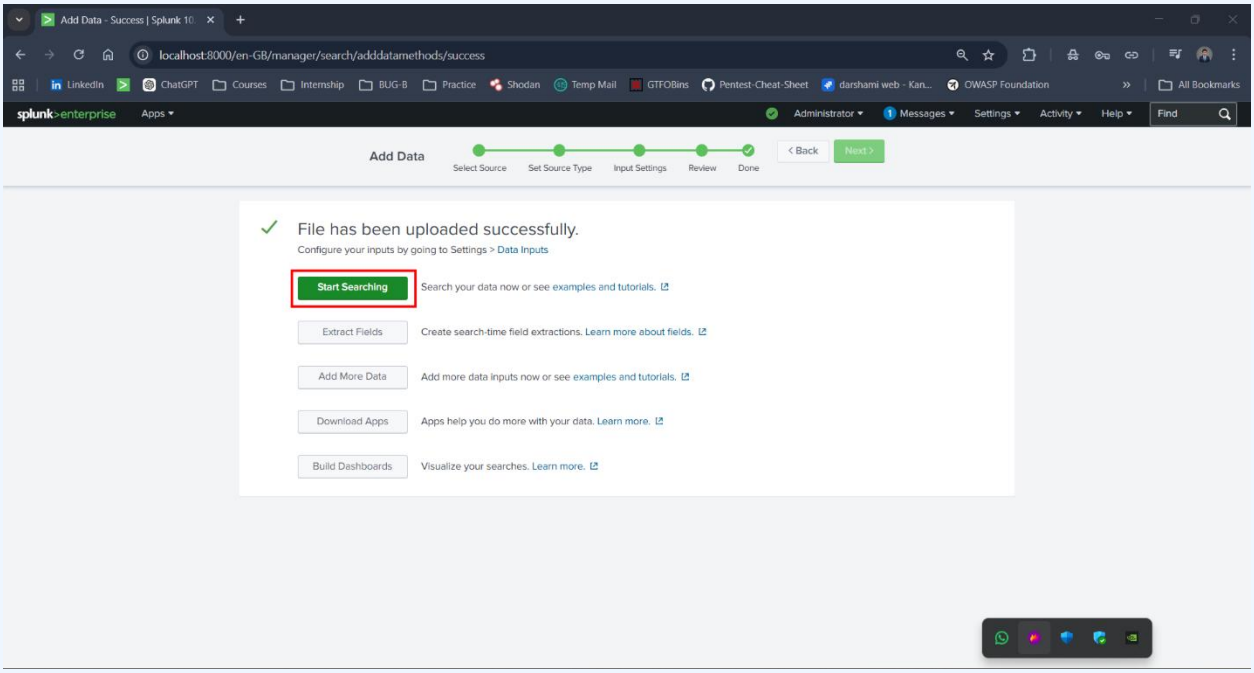
1. Review the configuration.
2. Click **Submit**.

To confirm indexing run:

```
index=http_lab
```

If events appear, the data ingestion was successful.





UNDERSTANDING HTTP LOGS

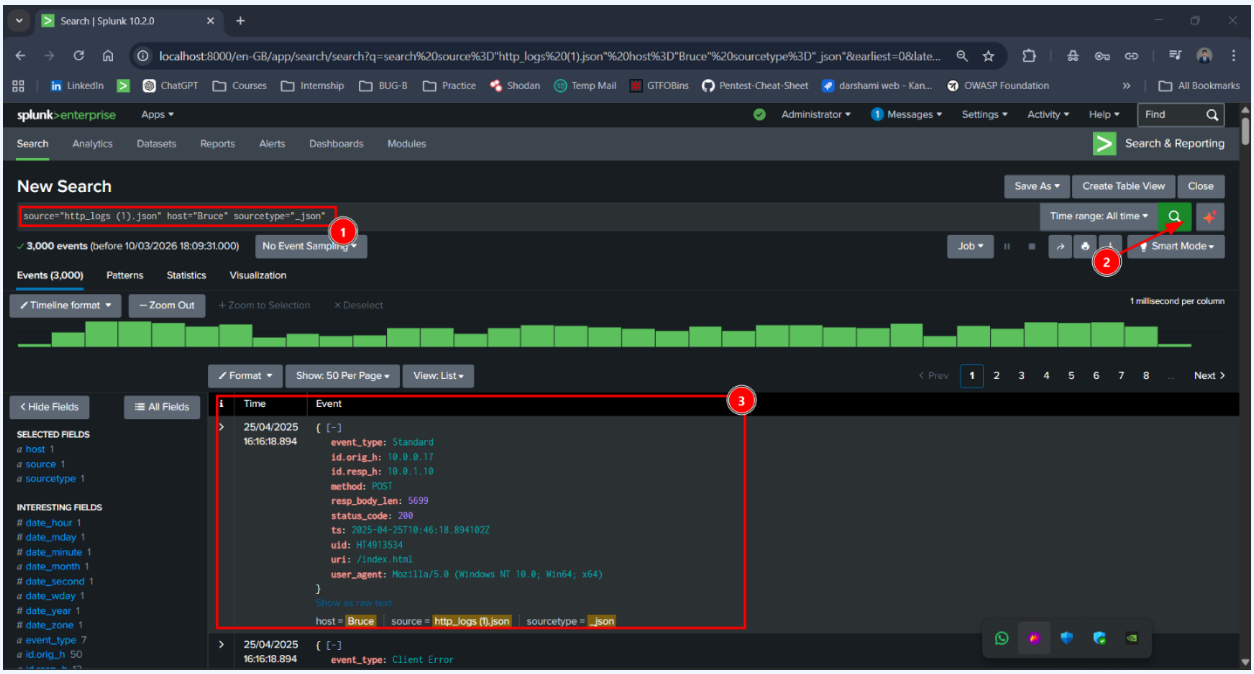
HTTP logs record **web requests and responses between clients and servers**.

Common fields in Zeek HTTP logs:

Field	Description
id.orig_h	Client IP address
id.resp_h	Web server IP
uri	Requested resource
user_agent	Client application making request
status_code	HTTP response status
resp_body_len	Size of server response

Example HTTP Log Event:

```
{
"id.orig_h":"192.168.1.10",
"id.resp_h":"192.168.1.5",
"uri":"/login.php",
"user_agent":"curl/7.68.0",
"status_code":200,
"resp_body_len":10240
}
```



LAB TASKS

Task 1: Find Top 10 Endpoints Generating Web Traffic

SPL Query

```
index=http_lab sourcetype="json"
| stats count by "id.orig_h"
| sort -count
| head 10
```

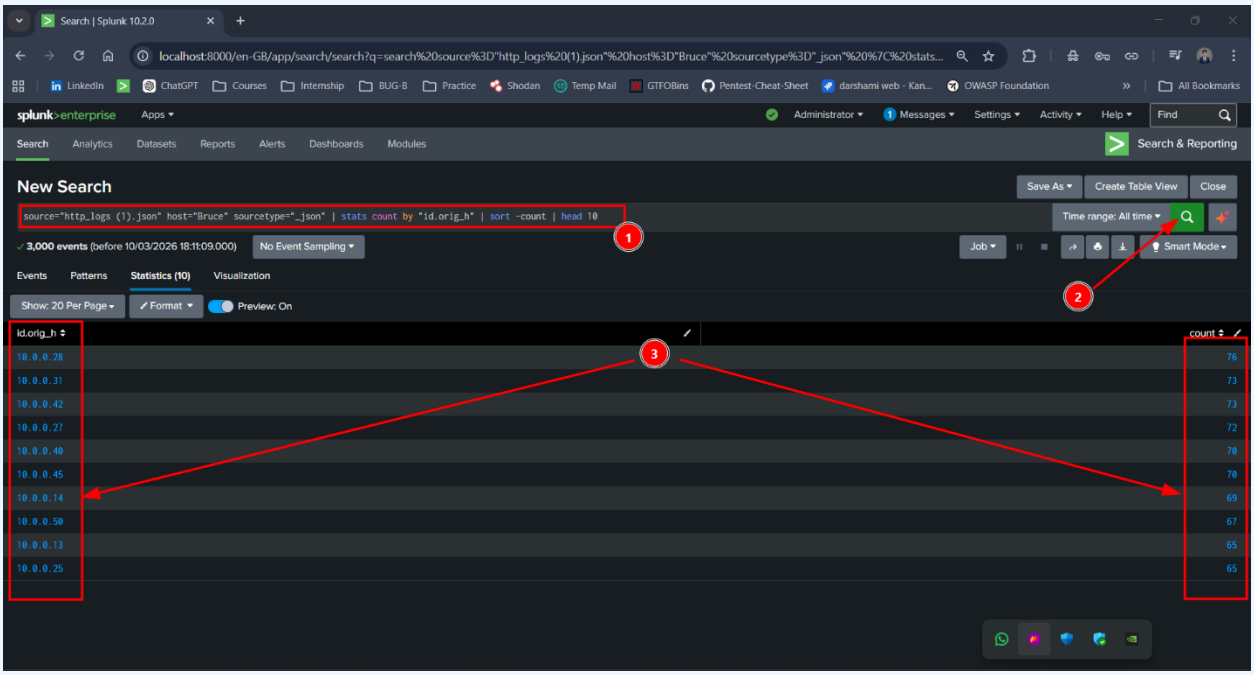
Explanation

Command	Purpose
index=http_lab	Searches logs stored in http_lab
sourcetype="json"	Filters JSON logs
stats count by id.orig_h	Counts HTTP requests from each client IP
sort -count	Sorts results in descending order
head 10	Displays the top 10 IP addresses

What This Shows

This query helps identify:

- Clients generating the most web traffic
- Potential scanning hosts
- Systems performing automated requests



Task 2: Count Server Errors (HTTP 5xx)

HTTP **5xx errors** indicate problems on the server side.

SPL Query

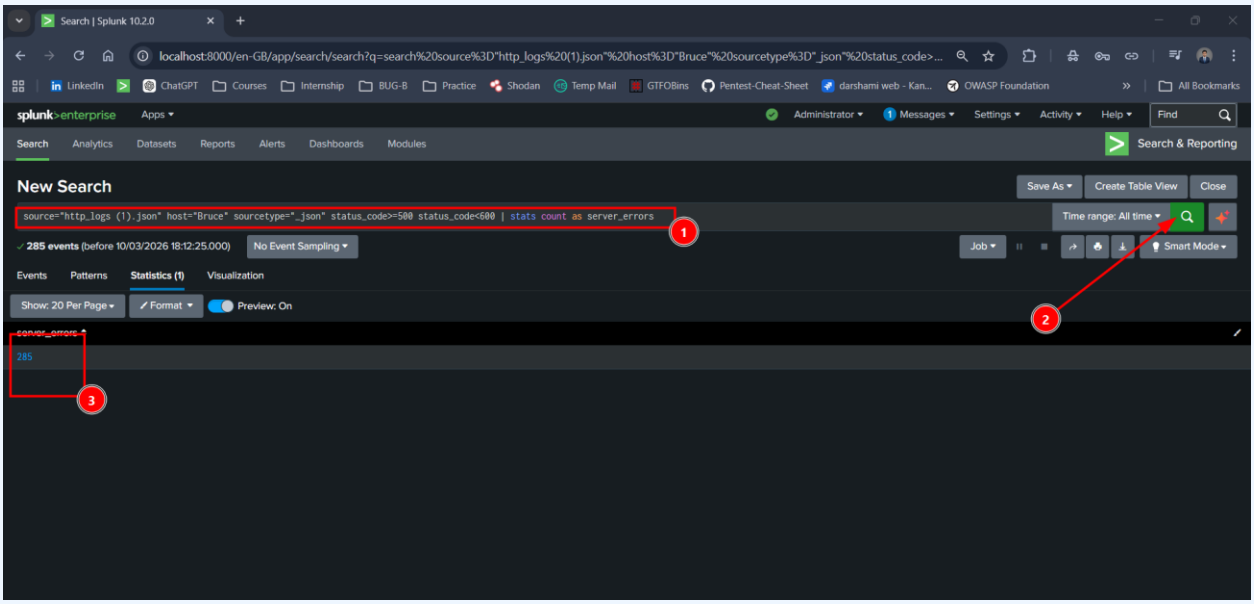
```
index=http_lab sourcetype="json" status_code>=500 status_code<600
| stats count as server_errors
```

Explanation

Command	Purpose
status_code>=500 status_code<600	Filters server error responses
stats count	Counts matching events
as server_errors	Renames result field

What This Shows

This query identifies **how many server-side errors occurred**.



Large numbers of 5xx errors may indicate:

- Web application issues
- Server overload
- Exploitation attempts

Task 3: Identify User Agents Associated with Scripted Attacks

Attack tools often use specific **user-agent strings**.

SPL Query

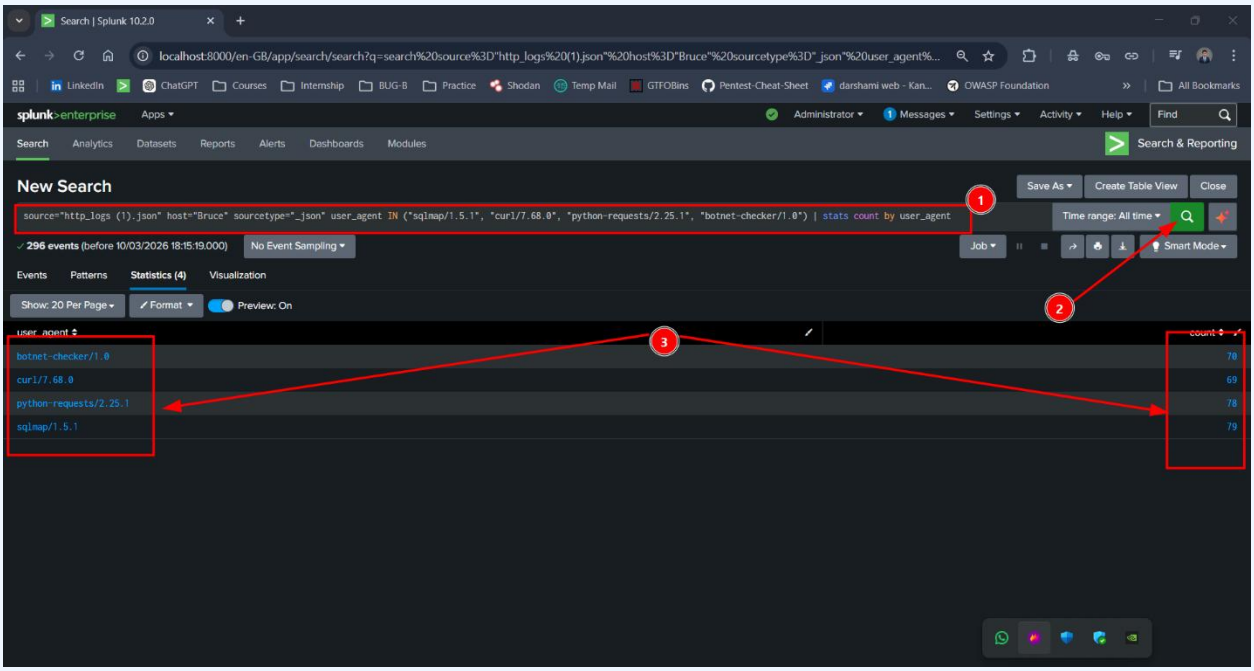
```
index=http_lab sourcetype="json" user_agent IN ("sqlmap/1.5.1",
"curl/7.68.0", "python-requests/2.25.1", "botnet-checker/1.0")
| stats count by user_agent
```

Explanation

Command	Purpose
user_agent IN (...)	Filters requests from known attack tools
stats count by user_agent	Counts occurrences per user agent

Known Suspicious Tools

User Agent	Description
sqlmap	SQL injection testing tool
curl	Command-line HTTP client
python-requests	Python automation library
botnet-checker	Possible botnet testing tool



These tools may indicate **automated scanning or exploitation attempts**.

Task 4: Identify Large File Transfers (>500 KB)

Large transfers may indicate **file downloads, uploads, or data exfiltration**.

SPL Query

```
index=http_lab sourcetype="json" resp_body_len>500000
| table ts "id.orig_h" "id.resp_h" uri resp_body_len
| sort -resp_body_len
```

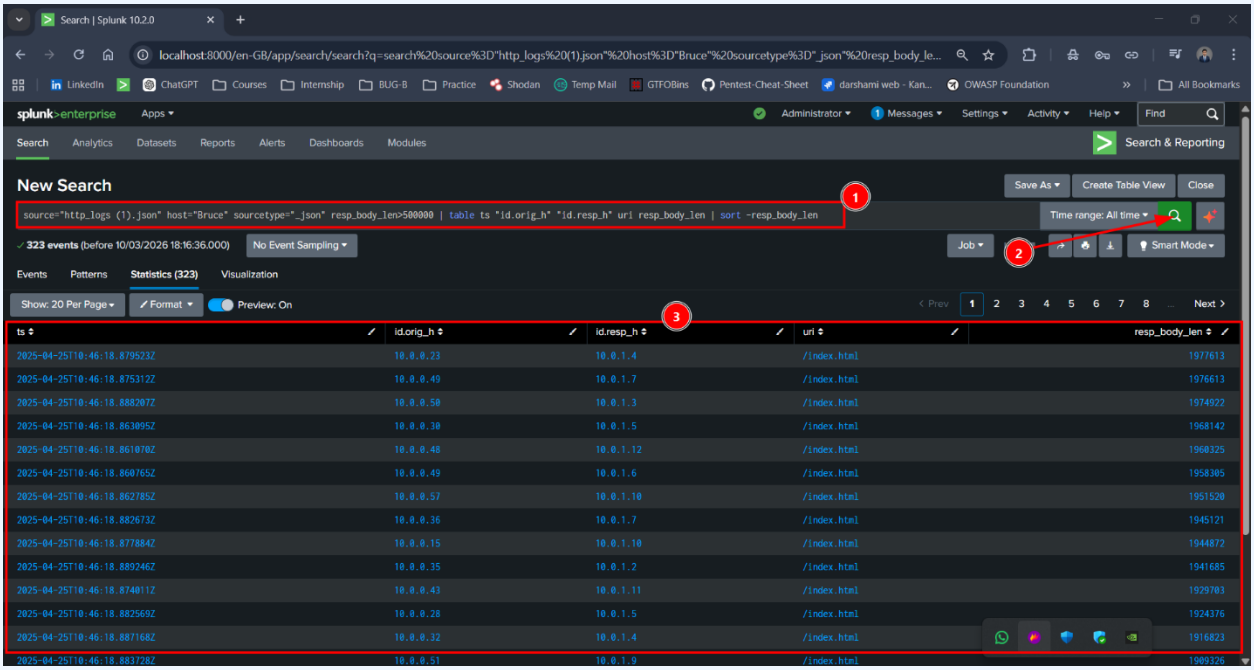
Explanation

Command	Purpose
resp_body_len>500000	Filters responses larger than 500 KB
table	Displays selected fields
sort -resp_body_len	Sorts results by largest transfer

What This Shows

This query identifies:

- Large downloads
- Possible data exfiltration
- File transfers over HTTP



Large transfers could indicate **file downloads or data leakage**.

SECURITY INSIGHTS

HTTP log analysis can reveal many types of attacks:

1. **Web Scanning:** Attack tools sending many automated requests.
2. **SQL Injection Testing:** Tools like **sqlmap** probing web applications.
3. **Botnet Communication:** Suspicious user agents connecting to servers.
4. **Data Exfiltration:** Large file transfers leaving the network.

Splunk enables **real-time monitoring and detection of these activities**.

CONCLUSION

In this lab we successfully:

- Uploaded **Zeek HTTP logs into Splunk**
- Learned how Splunk processes **JSON formatted web logs**
- Used **SPL queries to analyze HTTP traffic**

The analysis helped detect:

- Top clients generating web traffic
- Server-side errors
- Suspicious automated tools
- Large file transfers

These techniques are important for **web security monitoring, intrusion detection, and incident response**.